

Project Proposal

Group 30

Salma Baig, 400447531, baigs19

Yashika Yashika, xxxx, yashikay

Lucas Zheng, 400556902, zhengq42

October 9, 2026

COMPSCI/SFWRENG 4AL3: Applications of Machine Learning

Fall 2026

1 Overview

This project focuses on training a machine learning model to accurately segment brain tumors from MRI scans using a 2D U-Net convolutional neural network. Brain tumor classification involves identifying tumor regions within an MRI image.

Automated systems have the potential to support doctors by providing an initial segmentation that can be reviewed by the doctor, helping reduce the time for manual segmentation. Further, automated systems can in many cases act as a proof-reader, providing an additional safeguard towards any false-negatives.

Brain tumor classification is a challenging machine learning problem since tumors can vary significantly in appearance between subjects. Further, tumor tissue may also be similar visually to surrounding healthy tissue, or may only take up a small region of the MRI image.

2 Solution

(preliminary plan, subject to change)

Each MRI scan consists of four images: T1, contrast-enhanced T1 (T1CE), T2, and T2-FLAIR. These images will be stacked as separate input channels, allowing the model to use information from each image concurrently in its prediction.

The model will be built based off a U-Net architecture, which is a tried and proven workflow designed specifically for biomedical image classification tasks. PyTorch will be used for implementation and training/validation.